Scattering by a Central Potential; the Hamiltonian

15 approved segments, in reading order

Scattering by a central potential

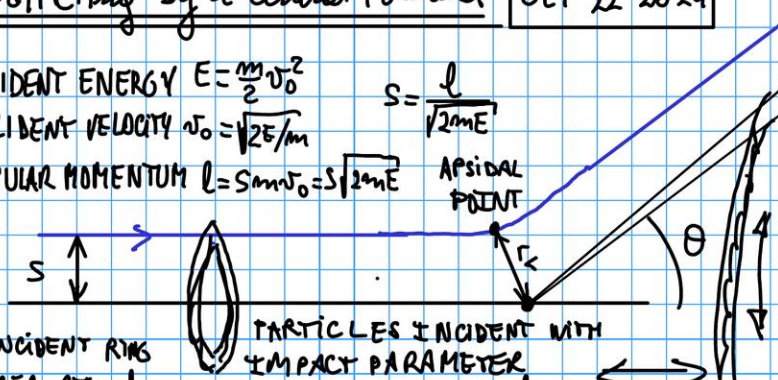
OCT 22 2024

INCIDENT ENERGY $E = \frac{m}{2} v_0^2$

INCIDENT VELOCITY $v_0 = \sqrt{2E/m}$

ANGULAR MOMENTUM $l = s m v_0 = s\sqrt{2mE}$

$s = \frac{l}{\sqrt{2mE}}$



INCIDENT RING AREA: $2\pi s\, ds$

PARTICLES INCIDENT WITH IMPACT PARAMETER BETWEEN s AND $s+ds$

$s \longleftrightarrow \theta$

$d^3r = r^2 dr \underbrace{d\varphi \sin\theta\, d\theta}_{d\Omega}$

SCATTERED RING SOLID ANGLE: $d\Omega = 2\pi \sin\theta\, d\theta$

Azimuthal angle φ irrelevant by symmetry

$s(\theta, E)$

PARTICLES EMERGING WITH SCATTERING ANGLE BETWEEN θ AND $\theta + d\theta$

INCIDENT INTENSITY

$$I = \frac{\#\text{ PART INCIDENT PER UNIT TIME}}{\text{INCIDENT BEAM CROSS SECTION AREA}}$$

DIFFERENTIAL CROSS SECTION

$$\sigma(\Omega)\, d\Omega = \frac{\#\text{PART. SCATTERED INTO } d\Omega \text{ PER UNIT TIME}}{\text{INCIDENT INTENSITY}}$$

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$$\sigma(\Omega)\,d\Omega = \frac{\#\text{PART. SCATTERED INTO } d\Omega \text{ PER UNIT TIME}}{\text{INCIDENT INTENSITY}}$$

$$I\,(2\pi s)\,|ds| = \#\text{ part going through ring between } s \text{ and } s+ds = \#\text{ part emerging with angle between } \theta \text{ and } \theta+d\theta = I\,\sigma(\theta)\,2\pi\sin\theta\,|d\theta|$$

THEREFORE:

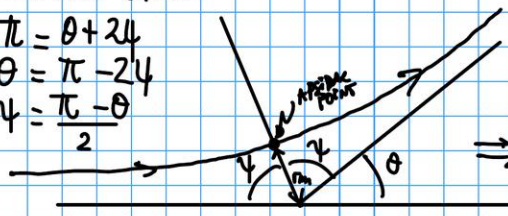
$$\boxed{\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{ds}{d\theta}\right|}$$

DEFINE ALSO THE TOTAL CROSS SECTION:

$$\boxed{\sigma_T = \int d\Omega\,\sigma(\Omega)}$$

REPULSIVE CASE:

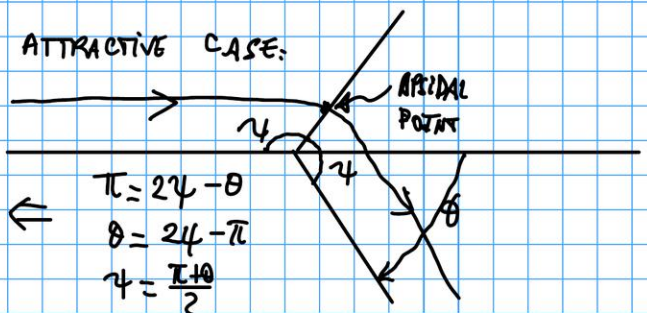
$$\pi = \theta + 2\psi$$
$$\theta = \pi - 2\psi$$
$$\psi = \frac{\pi - \theta}{2}$$



IN ALL CASES:

$$\Rightarrow \boxed{\theta = |\pi - 2\psi|} \Leftarrow$$

ATTRACTIVE CASE:



$$\pi = 2\psi - \theta$$
$$\theta = 2\psi - \pi$$
$$\psi = \frac{\pi + \theta}{2}$$

COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$$E = \frac{m}{2}\dot{r}^2 + V(r) + \frac{l^2}{2mr^2} \qquad \dot{r} = \left[\frac{2}{m}\left(E - V(r) - \frac{l^2}{2mr^2}\right)\right]^{1/2} \qquad \dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$$

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$$l = mr^2\dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{l}{mr^2} \Rightarrow d\theta = \dot{\theta}\,dt = \frac{l}{mr^2}dt = \frac{l}{mr^2}\frac{dr}{\left[\frac{2}{m}\left(E - V(r) - \frac{l^2}{2mr^2}\right)\right]^{1/2}} = \underbrace{\frac{l}{m}\sqrt{\frac{m}{2E}}}_{=\frac{l}{\sqrt{2mE}r^2}=s}\frac{r^{-2}\,dr}{\left[1 - \frac{V(r)}{E} - \underbrace{\frac{l^2}{2mE}}_{=s^2}\frac{1}{r^2}\right]^{1/2}}$$

$$\psi = \int_0^{\psi} d\psi = \int_{\theta_m}^{\pi} d\theta = \int_{r_m}^{\infty} \frac{s\,r^{-2}\,dr}{\left[1 - V(r) - s^2/r^2\right]^{1/2}} \Rightarrow \boxed{\theta(s) = \left|\pi - 2\int_0^{u_m} \frac{s\,du}{\left[1 - \frac{V(u)}{E} - s^2u^2\right]^{1/2}}\right|}$$

$u = 1/r \qquad du = r^{-2}\,dr \qquad u_m = 1/r_m$

From here, compute $\left|\frac{ds}{d\theta}\right| = \frac{1}{\left|\frac{d\theta}{ds}\right|}$ and use it to get $\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{ds}{d\theta}\right|$

UP TO HERE, RESULTS ARE VALID FOR <u>ANY</u> CENTRAL POTENTIAL $V(r)$

Coulomb Scattering

$E>0$	hyperbolic
$E=0$	parabolic
$E<0$	eliptic

ONLY FOR $V=-\frac{k}{r}$!!!

FORCE $f=\frac{Z Z' e^2}{r^2}$ $\quad Ze \circlearrowleft \quad Z'e \rightarrow$ $\quad f=-\frac{k}{r^2}$ $\quad V=-\frac{k}{r}$

$k=-Z Z' e^2$

Scattering: $E>0$

$$\frac{1}{r}=C\left[1+\epsilon\cos(\tilde{\theta}-\tilde{\theta}')\right] \qquad C=\frac{mk}{l^2}=-\frac{m}{l^2}Z Z' e^2$$

Eccentricity $\epsilon=\left[1+\frac{2El^2}{mk^2}\right]^{1/2}=\left[1+\frac{2El^2}{m(ZZ'e^2)^2}\right]^{1/2}=\left[1+\left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2}>1$

$l=s\,m v_0=s\sqrt{2mE} \rightarrow \frac{2l^2E}{m}=\frac{s^2 Z^2 Z'^2 m E^2}{m}=(2sE)^2$

eccentricity $\epsilon = \left[1 + \frac{2\ell^2 E}{m k^2}\right]^{\frac{1}{2}} = \left[1 + \frac{2\ell^2 E}{m(zz'e^2)^2}\right]^{\frac{1}{2}} = \left[1 + \left(\frac{2sE}{zz'e^2}\right)^2\right]^{\frac{1}{2}}$

$\ell = s m v_0 = s\sqrt{2mE} \rightarrow \frac{2\ell^2 E}{m} = \frac{s^2 2^2 m E^2}{m} = (2sE)^2$

2 SITUATIONS: i) REPULSIVE CASE $zz' > 0 \Rightarrow k = -zz'e^2 < 0$

ii) ATTRACTIVE CASE $zz' < 0 \Rightarrow k = -zz'e^2 > 0$ (ALSO GRAVITY $k = G m_1 m_2$)

i) REPULSIVE CASE: $k < 0$ $C_1 = \frac{mk}{\ell^2} < 0$

$\frac{1}{r} = C_1\left[1 + \epsilon\cos(\tilde{\theta} - \tilde{\theta}')\right] > 0 \Leftrightarrow 1 + \epsilon\cos(\tilde{\theta} - \tilde{\theta}') < 0 \Leftrightarrow \epsilon\cos(\tilde{\theta} - \tilde{\theta}') < -1 \Leftrightarrow \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\epsilon}$

We need the cosine to be negative and larger in magnitude than $1/\epsilon$

In this case, $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because $\cos(\tilde{\theta}' - \tilde{\theta}') = \cos(0) = 1 \Rightarrow \frac{1}{r} = C_1[1+\epsilon] < 0$ UNPHYSICAL

SO $\tilde{\theta} = \tilde{\theta}'$ IS IN THE MIDDLE OF THE INACCESSIBLE REGION OF THE HYPERBOLA, AND $\tilde{\theta} = \tilde{\theta}' \pm \pi$ WILL BE THE PERIAPSIS, SO WE CHOOSE $\tilde{\theta}' = \pi$ AND WE GET

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$$\cos(\tilde{\theta}-\tilde{\theta}') = \cos(\tilde{\theta}-\pi) = \cos\tilde{\theta}\underbrace{\cos\pi}_{-1} + \sin\tilde{\theta}\cancel{\sin\pi}^{0} = -\cos\tilde{\theta}$$

$$\frac{1}{r} = \left(-\frac{mk}{l^2}\right)\left[-1-\epsilon\cos(\tilde{\theta}-\tilde{\theta}')\right] = \frac{mZZ'e^2}{l^2}\left[\epsilon\cos\tilde{\theta}-1\right]$$

MAXIMUM AT $\tilde{\theta}=0 \Rightarrow$
MINIMUM r: $\frac{1}{r_m} = \frac{mZZ'e^2}{l^2}[\epsilon-1]$

SO THE ANGLE AT THE APSIDAL POINT IS $\tilde{\theta}=0$

WE NOW NEED $\psi = \tilde{\theta}(r\to\infty) = \tilde{\theta}\left(\frac{1}{r}\to 0\right)$

[IN THE REPULSIVE CASE $\theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi-\theta}{2}$]

$$0 = \epsilon\cos\psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos\psi = \cos\left(\frac{\pi-\theta}{2}\right) = \cancel{\cos\frac{\pi}{2}}^{0}\cos\frac{\theta}{2} + \underbrace{\sin\frac{\pi}{2}}_{1}\sin\frac{\theta}{2} = \sin\frac{\theta}{2}$$

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$\left[\ \theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi-\theta}{2}\ \right]$

ii) ATTRACTIVE CASE: $k>0 \quad G = \frac{km}{\ell^2} > 0$

TAKE $\tilde{\theta}' = 0 \quad \tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = G\left[1+\epsilon\cos(0)\right] = G\left[1+\epsilon\right]$

MAX AT $\tilde{\theta} = 0$

MINIMUM r: $\frac{1}{r_m} = G\left[1+\epsilon\right]$

APSIDAL POINT IS ALSO AT $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r\to\infty) = \tilde{\theta}\left(\frac{1}{r}\to 0\right)$

$$0 = 1 + \epsilon\cos\psi \iff -\frac{1}{\epsilon} = \cos\psi = \cos\left(\frac{\pi+\theta}{2}\right) = \underbrace{\cos\frac{\pi}{2}}_{0}\cos\frac{\theta}{2} - \underbrace{\sin\frac{\pi}{2}}_{1}\sin\frac{\theta}{2} = -\sin\frac{\theta}{2}$$

[IN THE ATTRACTIVE CASE $\theta = 2\psi - \pi \Rightarrow \psi = \frac{\pi+\theta}{2}$]

We now combine both cases:

$$\cot^2\frac{\theta}{2} = \frac{1}{\tan^2\frac{\theta}{2}} = \frac{\cos^2\theta/2}{\sin^2\theta/2} = \frac{1-\sin^2\frac{\theta}{2}}{\sin^2\frac{\theta}{2}} = \frac{1}{\sin^2\frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{zz'e^2}\right)^2 \Rightarrow \boxed{s = s(\theta,\epsilon) = \left|\frac{zz'e^2}{2E}\right| \cot\frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left|\frac{zz'e^2}{2\epsilon}\right| \frac{d}{d\theta}\left(\frac{\cos\theta/2}{\sin\theta/2}\right) = \left|\frac{zz'e^2}{2\epsilon}\right| \left(\frac{1}{2}\left(\frac{-\sin\theta/2}{\sin\theta/2}\right) - \frac{1}{2}\frac{\cos^2\theta/2}{\sin^2\theta/2}\right) = -\left|\frac{zz'e^2}{4E}\right| \frac{1}{\sin^2\frac{\theta}{2}}$$

$$\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{ds}{d\theta}\right| = \frac{1}{2}\left(\frac{zz'e^2}{2E}\right)^2 \frac{1}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} \frac{\cos\theta/2}{\sin\theta/2} \frac{1}{\sin^2\frac{\theta}{2}}$$

$$\boxed{\sigma(\theta) = \frac{1}{4}\left(\frac{zz'e^2}{2E}\right)^2 \csc^4\left(\frac{\theta}{2}\right)}$$

Rutherford scattering cross section (for Coulomb forces)

$$\sigma(\theta) = \frac{1}{4}\left(\frac{zz'e^2}{2E}\right)^2 \csc^4\left(\frac{\theta}{2}\right)$$

Rutherford scattering cross section (for Coulomb forces)

Remarks:

1) The cross section is the same for the repulsive and attractive cases, since it is proportional to (zz')^2.
2) It diverges like E^{-2} at low energies.
3) It diverges like \theta^{-4} (VERY STRONG!!!) for small angles \theta.
4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

Total scattering cross section:

$$\sigma_T = \int \sigma(\Omega)\, d\Omega$$

NOTATION:

Goldstein $\sigma(\Omega)$ $\leftrightarrow$ others $\frac{d\sigma}{d\Omega}$ DIFFERENTIAL SCATTERING CROSS SECTION

Total scattering cross section:

$$\sigma_T = \int \sigma(\Omega)\, d\Omega$$

In this case $\sigma(\theta) = \text{const} \cdot \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$

$$d\Omega = 2\pi \sin\theta\, d\theta = 4\pi \sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2}\, d\theta$$

$$\sigma_T = \frac{4\pi}{4}\left(\frac{z z' e^2}{2E}\right)^2 \int_0^{\pi} \frac{\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}}\, d\theta \sim \int_0^{\delta} \frac{d\theta}{(\theta/2)^3} = 8\int_0^{\delta} \frac{d\theta}{\theta^3} = 8\,\frac{\theta^{-2}}{(-2)}\Big|_0^{\delta} = +\infty$$

STRONGLY DIVERGENT

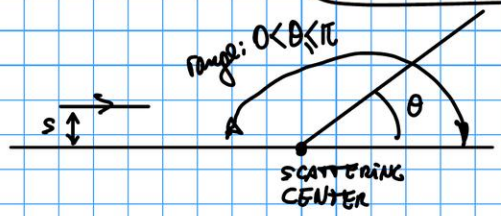
NOTATION:

Goldstein: $\sigma(\Omega)$ ↔ others: $\frac{d\sigma}{d\Omega}$ DIFFERENTIAL SCATTERING CROSS SECTION

σ_T ↔ σ TOTAL SCATTERING CROSS SECTION

$$\sigma_T = \frac{4\pi}{4}\left(\frac{zz'e^2}{2E}\right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}}\,d\theta \sim \int_0^\delta \frac{d\theta}{(\theta/2)^3} = 8\int_0^\delta \frac{d\theta}{\theta^3} = 8\,\frac{\theta^{-2}}{(-2)}\Big|_0^\delta = +\infty$$

STRONGLY DIVERGENT



CLASSICAL $\sigma_T = +\infty$ for any long range potential

QUANTUM $\sigma_T = +\infty$ for some long range potentials
$V(r)$ do not decay fast enough at $r\to\infty$

CAVEAT

1) IN A DENSE CONDUCTING MATERIAL: $V = -\frac{k}{r} \longrightarrow V^{\text{SCREEN}}(r) = -\frac{k}{r}\,e^{-r/\lambda}$ SCREENED COULOMB POTENTIAL ("YUKAWA")

(In non-conducting materials, $V^{\text{SCREEN}}(r)$ may take a different form, but still will vanish much faster than $\frac{1}{r}$)

Chapter 8 Hamiltonian and Hamilton's Eqns of Motion

$$L = T - V = L(q_1 \cdots q_n, \dot{q}_1 \cdots \dot{q}_n, t)$$

$$0 = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\bigg|_{\substack{q_{j\neq i} \\ \dot{q}_{j\neq i}}}\right) - \frac{\partial L}{\partial q_i}\bigg|_{\substack{q_{j\neq i} \\ \dot{q}_{j\neq i}}} \qquad p_i = \frac{\partial L}{\partial \dot{q}_i}\bigg|_{\substack{q_{j\neq i} \\ \dot{q}_{j\neq i}}}$$

Legendre Tranformation $q_1 \cdots q_n, \dot{q}_1 \cdots \dot{q}_n \rightarrow q_1 \cdots q_n, p_1 \cdots p_n$

$f(x,y) \Rightarrow df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_\theta = l_z$$

~~$L = L(r, \dot{r}, p_\theta)$ $\Downarrow$ Lagrange Eqn.~~

Legendre transformation $q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n \to q_1 \dots q_n, p_1 \dots p_n$

$f(x,y) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$

In Thermodynamics

1st Law $\quad dU = \cancel{dQ} + \cancel{dW} = T dS - p dV \quad$ (Fluid at constant N) $\quad U = U(S,V) \quad$ Internal Energy

We want T, V as variables

$F = A = U - TS$ Helmholtz Free Energy

$dA = dU - d(TS) = \cancel{T dS} - p dV - \cancel{T dS} - S dT = -S dT - p dV \Rightarrow A = A(T,V)$

$F = A = U - TS$ Helmholtz Free Energy

$dA = dU - d(TS) = \cancel{TdS} - pdV - \cancel{TdS} - SdT = -SdT - pdV \Rightarrow A = A(T,V)$

In general

$f(x,y) = u\,dx + v\,dy \qquad u = \frac{\partial f}{\partial x} \qquad v = \frac{\partial f}{\partial y} \qquad (x,y) \longrightarrow (u,y)$

$g = f - ux \Rightarrow dg = df - d(ux) = \cancel{u\,dx} + v\,dy - \cancel{u\,dx} - x\,du = -x\,du + v\,dy$

$g = g(u,y)$

Let's look at $L(q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n)$ $\qquad p_i = \left.\frac{\partial L}{\partial \dot{q}_i}\right|_{\substack{q_j \\ \dot{q}_{k\neq i}}}$

$H = -\left(L - \sum_i p_i \dot{q}_i\right) \qquad dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \cancel{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i}\right) + \sum_i \left(\cancel{p_i d\dot{q}_i} + \dot{q}_i dp_i\right) - \cancel{\partial L/\partial t} +$

$$H = -\left(L - \sum_i p_i \dot{q}_i\right) \qquad dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i\right) + \sum_i \left(p_i d\dot{q}_i + \dot{q}_i dp_i\right)$$

$$dH = -\frac{\partial L}{\partial t} + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i\right) \Rightarrow \quad H = H(p_1 \cdots p_n, q_1 \cdots q_n, t)$$

$$\frac{\partial H}{\partial p_i} = \dot{q}_i \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} \overset{\text{Lagrange Eqn}}{=} -\dot{p}_i \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

CANONICAL EQS OF MOTION